Rigid-Body Symmetry in Flow-Matching Motion Planning: What Canonicalisation Buys, and What It Does Not

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Abstract

This is a controlled study of how to exploit rigid-body symmetry in a generative motion planner, not a proposal for a competitive planner. Motion planning is exactly equivariant under rigid transformations, and a substantial literature builds that symmetry into network architectures. We hold the architecture, data and training budget fixed, vary only the representation of the problem, and find that almost all of the available benefit is obtained without touching the architecture at all — and we give a cheap measurement that says, *before* an equivariant model is built, how much benefit is left for it to capture.

Our starting point is that the start and goal of a planning query themselves define a frame, and that this frame removes five of the six degrees of freedom of SE(3) exactly, at zero computational cost. The sixth — a roll about the start–goal axis — cannot be removed by any continuous rule; we give two independent obstructions, one topological and one distributional. We therefore treat the sixth by symmetry: roll augmentation during training and frame averaging at inference.

The size of the effect is best read against a floor rather than in isolation. On a cluttered 3D point-mass benchmark, world-frame flow matching reaches 15.4% of individual samples collision-free on held-out environments — statistically indistinguishable from the 15.6% obtained by joining start to goal with a straight line, measured on the identical problems. Per sample, training on raw world coordinates buys nothing over joining the endpoints with a line. With the reduction, the same network, data and budget reach 45.6%, and, contrary to the hypothesis that canonicalisation is a small-data crutch, the gap *widens* monotonically with training data: the world-frame arm gains 2.6 points across a 12.5× increase in training environments while the canonicalised arm gains 26.6 and has not plateaued.

The two mechanisms built for the sixth degree of freedom contribute +1.2 and +0.2 points, both inside noise. We explain this rather than merely reporting it, by measuring the *non-equivariance residual*: only 1.7% of the learned velocity field lies outside the equivariant subspace. Because frame averaging is the orthogonal projection onto that subspace, 1.7% is the entire budget any post-hoc symmetrisation of this model can spend. We propose the residual as a cheap diagnostic of how much non-equivariant structure a learned field retains: it is one forward pass, needs no retraining, and it quantifies the headroom available to post-hoc symmetrisation exactly. It does *not* forecast what a symmetry mechanism is worth, and we show this rather than assume it: measured under the full SE(3) action, augmentation and canonicalisation leave the field equally equivariant ($r = 0.018$ against 0.017) while differing by 28 points of held-out performance. Learning a symmetry and solving the problem are separable, and r measures only the first.

The obvious alternative — exploiting the same symmetry by augmenting training with random rigid motions — recovers little of this: at 250 environments it lifts the world-frame arm from 15.4% to 17.5%, about 7% of what the reduction achieves. We also find that optimisation and representation are *separate* bottlenecks: tripling the training budget is worth +8.0 points to the canonicalised arm and +0.9 to the world-frame one, so compute cannot buy what the representation forecloses. We calibrate both ends on the same 500 problems: an untrained Brownian-bridge prior, swept over widths, never exceeds 15.8% at best-of-20, while classical planners reach 74–100% in under 0.4 s on one CPU core. The learned sampler does not approach the latter, and we present this as a controlled study of representation choices, not as a planner proposal.

I Introduction

A motion planning problem consists of a workspace populated with obstacles, a start configuration $\mathbf{s}$, and a goal configuration $\mathbf{g}$; its solution is a path τ joining them and avoiding the obstacles. The problem carries an exact symmetry: for any rigid transformation $T \in \mathrm{SE}(3)$,

$$\tau\big(T\cdot\mathcal{O},\, T\mathbf{s},\, T\mathbf{g}\big) \;=\; T\cdot\tau\big(\mathcal{O},\, \mathbf{s},\, \mathbf{g}\big), \qquad (1)$$

where $\mathcal{O}$ denotes the obstacle set. Nothing about the geometry changes when the problem is picked up and moved.

A neural planner trained on raw world coordinates has no access to Eq. (1). As far as its weights are concerned, planning left-to-right near the ceiling is an unrelated task from planning front-to-back near the floor, and the same skill must be relearned at every pose. The dominant response in the literature is architectural: constrain the layers so that equivariance holds by construction [5, 26, 9, 6, 10], an approach that has been carried into robot learning with real success [23, 28, 25, 29]. That success comes at a price that is rarely quantified: constrained layers cost expressiveness, are laborious to implement, and fail in ways that are silent.

There is a second route, older and much cheaper: build the symmetry into the *representation of the problem*, so that the network is never shown the pose and therefore cannot be sensitive to it [20, 13]. This paper asks how far that route goes on a generative motion planning problem, and — more importantly — how one can *know in advance* whether the expensive route is worth taking.

Our answer has four parts.

(i) Five of six degrees of freedom are free. The query supplies a frame. Placing the origin at the midpoint of $\mathbf{s}$ and $\mathbf{g}$ and aligning the first axis with the start–goal direction removes three translational and two rotational degrees of freedom, exactly and at no cost. The resulting representation is invariant under any rigid transformation of the problem *up to a single roll about the start–goal axis*. The quantities a roll leaves alone — the separation d, the first coordinate of every reduced point, and its radius in the plane normal to that axis — are invariant outright (Proposition 1). Within those five that is a stronger guarantee than a trained equivariant architecture provides, since it holds to floating-point precision at initialisation rather than to optimisation tolerance. The reduction therefore collapses SE(3) to SO(2); the remaining roll is what the rest of the paper is about. As a side effect the six numbers previously used to condition the network on the query collapse to a single invariant scalar.

(ii) The sixth is provably not removable, for two unrelated reasons. No continuous rule can fix the roll about the start–goal axis: such a rule is a nowhere-vanishing tangent field on S^2, which the hairy ball theorem forbids (Proposition 2). Deriving the roll from the scene instead also fails, because the obstacle distribution is symmetric under central inversion, so the first angular moment — the one a scene-derived gauge would actually use — vanishes in expectation for every start–goal direction (Proposition 3). We therefore handle the sixth degree of freedom by symmetry rather than canonicalisation.

(iii) The symmetry machinery is inert here, and we measure why. Roll augmentation and frame averaging contribute +1.2 and +0.2 percentage points, both inside noise, while the reduction contributes +30.3. Rather than infer inertness from flat curves — a weak conclusion — we measure the quantity frame averaging exists to remove. The *non-equivariance residual*, the disagreement among rotated copies of the velocity field, is 1.7% and is flat across a 12.5× range of training data. Since frame averaging is the orthogonal projection onto the equivariant subspace (Proposition 4), the error it can remove is bounded by that 1.7% (Corollary 5), while the model is failing on most of its samples: the model's error is overwhelmingly equivariant error. The residual is also smaller at the larger training scale (1.86% → 1.57%), which is consistent with the property arriving on its own, though two points at one seed do not establish a trend and we do not rest the argument on it.

(iv) Calibration against floors and ceilings. A collision-free rate is uninterpretable on a new benchmark, so we measure, on the identical 500 held-out problems, both a trivial lower bound and four classical planners. The trivial bound is the finding: a straight line from start to goal is collision-free on 15.6% of problems, and the world-frame model scores 15.4% per sample. The classical planners (74–100%, all under 0.4 s on a single CPU core) bound the other side and show that the learned sampler, at 45.6%, is not competitive with them here. Because a generative planner is deployed as best-of-N behind a collision check, we also report that per-query metric against the honest control for it — the model's own untrained prior, sampled N times, at its best width. That comparison separates two things the per-sample number conflates: the world-frame arm is at the straight-line floor per sample, but well above the untrained prior per query, so it has learned sample *diversity* without learning sample *quality*.

We regard (iii) as the transferable contribution. The residual is a few lines of code, costs one forward pass, requires no retraining, and can be computed on any model with a group action on its inputs. It converts "we tried equivariance and it did not help" — an anecdote — into a quantitative statement about how much headroom *post-hoc symmetrisation* has on a given problem, with an exact bound attached. We recommend it as a diagnostic to run before reaching for symmetrisation, and we are explicit about its limit: measured across three arms of this benchmark, r does not predict what a symmetry mechanism is worth. Two models with indistinguishable residuals differ by 28 points (Table 1), because one of them fixed the representation and the other only learned the symmetry.

II Related Work

Equivariant architectures. Group-equivariant convolution [5] generalises translation weight-sharing to larger groups; for SO(3) and SE(3) the standard constructions are tensor field networks [26], SE(3)-Transformers [9], vector neurons [6], and the `e3nn` framework [10]; [2] surveys the area. In robot learning these ideas underpin neural descriptor fields [25], equivariant descriptor fields [23], equivariant diffusion policy [28], and SIM(3)-equivariant policies [29]. This work does not dispute those results; it provides a measurement that indicates when they are the right instrument.

Canonicalisation and frame averaging. Frame averaging [20] obtains equivariance by averaging an unconstrained network over a set of frames derived from the input, and learned canonicalisation [13] predicts a single frame with a small auxiliary network. Both leave the backbone unconstrained. Our $(\mathbf{s}, \mathbf{g})$ reduction is a canonicalisation in this sense, with the distinguishing feature that the frame is not learned or searched but read directly off the query, so five degrees of freedom are removed in closed form; the obstruction of Proposition 2 is precisely the reason the sixth cannot be handled the same way, and is why we fall back to frame averaging there.

Generative motion planning. Diffusion and flow models over whole trajectories have become a standard planning substrate [12, 4, 3, 27]. We use the conditional flow matching / rectified flow formulation [15, 16, 1] for its straight-line interpolants and few-step sampling. Learned planners that regress directly to paths [22, 8] form the non-generative counterpart. Within this planning literature we are not aware of a reported measurement of how far the learned field departs from equivariance.

Measuring learned equivariance. The idea of measuring, rather than assuming, how equivariant a trained network is, is not new, and the closest work to ours is outside planning. Gruver et al. [11] measure learned equivariance through the Lie derivative of a trained model with respect to the group action, and find that unconstrained architectures often acquire substantial equivariance from data alone — the same qualitative phenomenon our residual reports, in a different parameterisation and for continuous groups. Elesedy and Zaidi [7] analyse the generalisation benefit of equivariance using the orthogonal decomposition of a function into equivariant and anti-equivariant parts, which is the decomposition Corollary 5 is stated in. Relative to these, what we add is narrow and practical: the residual is a single forward pass with no differentiation and no retraining, and it comes with an exact statement of what it bounds. We are careful not to claim more: it measures the headroom for symmetrising a *given* model, which is not the same question as how much a constrained architecture would gain.

Classical planners supply our supervision: RRT-Connect [14] for feasibility, then shortcutting and CHOMP [30] refinement, with TrajOpt [24] available in the benchmark.

III Problem Setup

A Benchmark

Experiments use PointMass3D, a benchmark we built for this study. A spherical robot of radius 0.03 moves in the workspace $[-1,1]^3$ populated with 20 spheres and 20 axis-aligned boxes per environment, obstacle centres drawn uniformly in $[-0.75, 0.75]^3$. Collision checking uses an analytic signed distance field: the environment SDF is the minimum over obstacle SDFs and the workspace walls, and clearance subtracts the robot radius. Start and goal are drawn uniformly subject to a collision margin and a minimum separation $\|\mathbf{g}-\mathbf{s}\| \geq 1.5$. Forty obstacles in a two-unit cube is dense clutter; absolute success rates are correspondingly low and should be read comparatively.

Expert demonstrations come from RRT-Connect, random shortcutting, uniform resampling to 64 waypoints, and CHOMP refinement, with dense collision validation and fallback to the raw RRT path. The dataset is 300 environments $\times$ 600 start–goal pairs $\times$ 30 trajectories, doubled by time-reversal.

B Flow-matching planner

The planner is a conditional flow-matching model over whole trajectories. A trajectory is $N = 64$ waypoints in $\mathbb{R}^3$. Training draws $\mathbf{x}_0 \sim \mathcal{N}(0, I)$ and $t \sim \mathcal{U}[0,1]$, forms the interpolant $\mathbf{x}_t = (1-t)\mathbf{x}_0 + t\,\mathbf{x}_1$ with $\mathbf{x}_1$ an expert trajectory, and regresses the velocity target:

$$\mathcal{L} = \mathbb{E}_{\mathbf{x}_0,\mathbf{x}_1,t}\big\|v_\theta(\mathbf{x}_t, t, c) - (\mathbf{x}_1 - \mathbf{x}_0)\big\|^2, \qquad (2)$$

where c encodes the obstacles and the query. Sampling integrates $\dot{\mathbf{x}} = v_\theta(\mathbf{x}, t, c)$ from $t = 0$ to 1 by explicit Euler. The network is a dilated temporal convolutional trunk with FiLM conditioning [18, 19] fed by a PointNet-style permutation-invariant obstacle encoder [21]; 2.16M parameters throughout.

IV Method

A Decomposing the group

SE(3) has six degrees of freedom. The query supplies a frame, and that frame consumes five of them:

- **Three translations**, removed by placing the origin at the midpoint $\mathbf{o} = \frac{1}{2}(\mathbf{s}+\mathbf{g})$.
- **Two rotations**, removed by aligning the first axis with $\hat{\mathbf{x}} = (\mathbf{g}-\mathbf{s})/\|\mathbf{g}-\mathbf{s}\|$. Specifying a direction on S^2 costs exactly two degrees of freedom.
- **One rotation remains**: the roll about $\hat{\mathbf{x}}$. Having fixed which way the path runs, nothing in the problem specifies which way is "up" around that axis.

Concretely, let $d = \|\mathbf{g}-\mathbf{s}\|$, let $i^* = \arg\min_i |\hat{\mathbf{x}}_i|$, and set

$$\hat{\mathbf{y}} \propto \hat{\mathbf{x}} \times \mathbf{e}_{i^*}, \qquad \hat{\mathbf{z}} = \hat{\mathbf{x}} \times \hat{\mathbf{y}}, \qquad (3)$$

with R the rotation whose rows are $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$. A point $\mathbf{p}$ maps to $R(\mathbf{p}-\mathbf{o})$. Because the smallest component of a unit vector satisfies $|\hat{\mathbf{x}}_{i^*}| \leq 1/\sqrt{3}$, the cross product in Eq. (3) has norm $\geq \sqrt{2/3} \approx 0.816$ everywhere, so the construction is well-conditioned over the entire domain with no threshold or fallback branch, and $\det R = +1$ by construction.

Proposition 1 (SE(3) reduces to a single roll)**.** *Let X denote the reduced representation of $(\mathcal{O}, \mathbf{s}, \mathbf{g})$ and X' that of $(T\mathcal{O}, T\mathbf{s}, T\mathbf{g})$ for $T = (Q, \mathbf{t}) \in \mathrm{SE}(3)$. Then there is a single angle $\theta(T, \mathbf{s}, \mathbf{g})$ such that*

$$X' \;=\; R_{\hat{\mathbf{x}}}(\theta)\, X, \qquad (4)$$

where $R_{\hat{\mathbf{x}}}(\theta)$ is the rotation by θ about the first axis, applied to every reduced quantity — waypoints, obstacle centres and obstacle edge vectors alike. Consequently d, the first coordinate of every reduced point, and its radius in the plane normal to that axis are invariant under all of SE(3)*; the residual ambiguity is exactly one degree of freedom.*

Proof sketch. Under T the midpoint maps to $Q\mathbf{o}+\mathbf{t}$ and the direction to $Q\hat{\mathbf{x}}$, so both frames send the start–goal direction to the first basis vector. A change of basis between two right-handed orthonormal frames that agree on their first axis is a rotation about that axis, which gives Eq. (4); the invariants named are exactly the invariants of that rotation. $\square$

It is worth being precise about what this does *not* say, because the stronger claim is tempting and false. The reduced representation is not invariant under SE(3). The gauge $\hat{\mathbf{y}}$ in Eq. (3) is built from a *fixed world axis* $\mathbf{e}_{i^*}$, so rotating the whole problem changes which axis is selected and rolls the frame. What Proposition 1 establishes is that the entire residual is that one roll — SE(3) collapses to SO(2), not to nothing. Five degrees of freedom are removed exactly, to floating-point precision, at initialisation and without any training; the sixth survives as a gauge ambiguity. This is precisely why the mechanisms of Sec. IV-C exist, and a reader who took the representation to be fully invariant would find those mechanisms unmotivated. We verify Eq. (4) numerically to 10^{-15} in both domains.

Two further consequences follow. The six numbers $(\mathbf{s},\mathbf{g})$ collapse to the single invariant scalar d, and every pair of training examples that were "the same problem in a different pose" now coincide up to a roll, raising the effective density of the training set without generating a single new trajectory.

B Why the sixth degree of freedom is not removable

Proposition 2 (Topological obstruction)**.** *There is no continuous map $\hat{\mathbf{y}} : S^2 \to S^2$ with $\hat{\mathbf{y}}(\hat{\mathbf{x}}) \perp \hat{\mathbf{x}}$ for all $\hat{\mathbf{x}}$. More generally, for a gauge $\hat{\mathbf{y}}(\hat{\mathbf{x}},\mathcal{O})$ that may also depend on the scene, fix any $\mathcal{O}$: then $\hat{\mathbf{x}} \mapsto \hat{\mathbf{y}}(\hat{\mathbf{x}},\mathcal{O})$ is a tangent field on S^2, and if continuous it vanishes somewhere.*

Proof. Such a map is a continuous nowhere-vanishing unit tangent vector field on S^2, whose existence the hairy ball theorem denies [17]. The second statement is the first applied to the section at fixed $\mathcal{O}$. $\square$

The second form is what the scene-derived gauges of Sec. IV-B need, and it is worth being precise about what it gives. Such a gauge is a function of $(\hat{\mathbf{x}},\mathcal{O})$, so the bare hairy ball theorem — a statement about maps out of S^2 alone — does not apply to it. Fixing the scene repairs that, at the cost of a weaker conclusion: the degenerate set is *per scene*. Every scene admits start–goal directions at which its gauge breaks down, but which directions those are moves as the scene moves, so no restriction of the query distribution can avoid them uniformly.

Every deterministic gauge therefore carries a discontinuity; one may relocate it but never remove it. The escape route of restricting to a contractible subdomain is closed here: the minimum separation 1.5 along an axis of extent 2.0 still admits axis-aligned start–goal directions, so the domain is the whole sphere. Our i^* tie-break is exactly such a relocated discontinuity, placed on a measure-zero set.

Proposition 3 (Distributional obstruction)**.** *Let the obstacle-centre law be invariant under the central inversion $\mathbf{p} \mapsto -\mathbf{p}$, let φ_j be the angles of the centres in the plane normal to $\hat{\mathbf{x}}$, and let the weights w_j depend only on radius. Then the angular moments $c_m = \sum_j w_j e^{im\varphi_j}$ satisfy*

$$\mathbb{E}[c_m] = 0 \quad \textit{for every odd } m \textit{ and every } \hat{\mathbf{x}} \in S^2. \qquad (5)$$

Proof. Inversion fixes every radius and hence every w_j, and maps $\varphi_j \mapsto \varphi_j + \pi$, so $e^{im\varphi_j} \mapsto (-1)^m e^{im\varphi_j}$ and $\mathbb{E}[c_m] = -\mathbb{E}[c_m]$ for odd m. $\square$

The hypothesis is deliberately weaker than our benchmark: uniform-on-a-cube is one centrally symmetric law, and the proof covers Gaussian clutter, spherical shells, and any symmetric mixture equally. Nothing here is a theorem about our own generator.

The restriction to odd m is not a weakness of the proof but a fact about the geometry, and it is worth stating because the natural stronger claim is false. One is tempted to argue that the cube is C_4-symmetric, so only multiples of four survive. That holds when $\hat{\mathbf{x}}$ is a four-fold axis of the cube, and fails otherwise, because the plane normal to a general $\hat{\mathbf{x}}$ is not a coordinate plane and the cube's symmetry group does not act on it as C_4. Numerically, at 2×10^5 scenes (noise floor 0.002), $|\mathbb{E}[c_2]|$ is 0.001 along a coordinate axis but 0.033 for a generic direction and 0.070 along a face diagonal, while $|\mathbb{E}[c_1]|$ is at the noise floor for all of them. Eq. (5) is what the geometry actually supports.

Eq. (5) is an *unconditional* statement, and the gauge is used conditionally — a point that matters, and that closes the route more firmly than the proposition alone. A scene-derived gauge is computed in the reduced frame, i.e. from the displacements $\mathbf{p}_j - \mathbf{o}$ with $\mathbf{o} = \frac{1}{2}(\mathbf{s}+\mathbf{g})$. Conditional on the query, the law of $\mathbf{p}_j - \mathbf{o}$ is the obstacle law shifted by $-\mathbf{o}$, which is centrally symmetric about $-\mathbf{o}$ rather than about the origin. So

$$\mathbb{E}[c_1 \mid \mathbf{s},\mathbf{g}] \;\neq\; 0 \quad \text{whenever} \quad \mathbf{o} \neq \mathbf{0}, \qquad (6)$$

with the surviving moment pointing along the projection of $-\mathbf{o}$ onto the plane normal to $\hat{\mathbf{x}}$. Eq. (5) recovers $\mathbb{E}[c_1] = 0$ only after averaging over queries, since $\mathbb{E}[\mathbf{o}] = \mathbf{0}$. We confirm Eq. (6) numerically: at $\mathbf{o} = (0, 0.4, 0.3)$ the conditional first moment has magnitude 0.56 and argument 2.208, against a predicted 2.214.

That is the sharper closing of the scene-derived route. Conditioned on a query with $\mathbf{o} \neq \mathbf{0}$ the first-moment gauge is *not* degenerate — it is well conditioned, and it points from the query midpoint towards the centre of the workspace. But that direction is not an SE(3)-invariant of the problem: it is a function of where the query sits in the world. A gauge built on it would reintroduce exactly the world-frame dependence Proposition 1 removes, forfeiting five degrees of freedom to fix the sixth. The first-moment gauge is therefore either degenerate (unconditionally, at $\mathbf{o} = \mathbf{0}$) or not equivariant (conditionally, elsewhere), and neither is usable.

A gauge built from an even moment escapes Eq. (5) — inversion does not kill even m — and also escapes Eq. (6), since an $m = 2$ moment is insensitive to the sign of the displacement. It does not escape Proposition 2 in its second form: an $m = 2$ moment defines a *director* rather than a vector, and a continuous director field on S^2 is obstructed for the same Euler-characteristic reason as a vector field, since $\chi(S^2) = 2 \neq 0$. The conclusion is again per scene: for each fixed $\mathcal{O}$ there are directions at which the director degenerates. Both routes to a canonical roll are closed, for unrelated reasons.

C Handling the sixth by symmetry

Roll augmentation. During training a uniform $\theta \sim \mathcal{U}[0, 2\pi)$ is composed into R, so reduction and roll are applied as a single rotation. The prior $\mathbf{x}_0$ is drawn *after* this rotation and never rotated: an isotropic Gaussian is already rotation-invariant, so rotating it is a redundant step capable only of introducing error.

Frame averaging. At inference the field is evaluated at K rolled copies of the state, each result rotated back, and averaged:

$$(\mathcal{A}f)(\mathbf{x}) = \frac{1}{K}\sum_{k=0}^{K-1} \rho(\theta_k)^{-1} f\big(\rho(\theta_k)\mathbf{x}\big), \quad \theta_k = \frac{2\pi k}{K}. \quad (7)$$

Proposition 4 (Exact C_K equivariance and projection)**.** *For any f, $\mathcal{A}f$ is exactly C_K-equivariant. Moreover $\mathcal{A}$ is linear, idempotent, and self-adjoint with respect to $\langle f, g\rangle = \mathbb{E}_{\mathbf{x}}\left[f(\mathbf{x})^\top g(\mathbf{x})\right]$ for any C_K-invariant law on $\mathbf{x}$; hence it is the orthogonal projection onto the subspace of C_K-equivariant fields.*

Proof sketch. Rolling the input by θ_j permutes the quadrature points, and reindexing a finite sum is a bijection on them, giving equivariance and $\mathcal{A}^2 = \mathcal{A}$. Self-adjointness follows by substituting $\mathbf{x} \mapsto \rho(\theta_k)\mathbf{x}$ in the expectation, which the invariance of the law leaves unchanged, together with $\rho(\theta_k)^\top = \rho(\theta_k)^{-1}$. □

We claim no novelty for Proposition 4 or for Corollary 5 below. That group averaging is the orthogonal projection onto the equivariant subspace, and that a function decomposes orthogonally into equivariant and anti-equivariant parts, are established results in the analysis of equivariant learning [7, 20]; we restate them because the bound we are proposing is read off them. The contribution here is the measurement — the size of the anti-equivariant component of a trained planning field, and what it implies for the mechanisms built to remove it — rather than the projection theorem.

Two properties of Eq. (7) matter in practice. The guarantee requires no architectural constraint and no training assumption — it is a property of the operator, and holds for an untrained network. And because the obstacle encoding is time-independent, the K rolled scene encodings are computed once outside the ODE loop and reused at every integration step, which is why $K{=}9$ costs 1.05× the wall-clock time of $K{=}1$ (Sec. V-F).

D The non-equivariance residual

Proposition 4 turns a qualitative question — "is symmetry worth enforcing here?" — into a measurable one. Write $v_k = \rho_k^{-1} f(\rho_k \mathbf{x})$ and $\bar{v} = \mathcal{A}f(\mathbf{x})$, and define

$$r = \frac{\left(\mathbb{E}_{\mathbf{x},k}\left\|v_k - \bar{v}\right\|^2\right)^{1/2}}{\left(\mathbb{E}_{\mathbf{x}}\left\|\bar{v}\right\|^2\right)^{1/2}}. \quad (8)$$

This is the relative norm of the component of the learned field that lies outside the equivariant subspace. The norms must be the Hilbert norms of Proposition 4, i.e. root-mean-square rather than mean-of-norms; Jensen's inequality separates the two, and only the form in Eq. (8) makes Corollary 5 an identity rather than an approximation.

Corollary 5 (Symmetrisation budget)**.** *Let f^* be the target velocity field, which is C_K-equivariant because the planning problem is. Since $\mathcal{A}$ is the orthogonal projection onto the equivariant subspace and $\mathcal{A}f^* = f^*$, the error splits orthogonally:*

$$\|f - f^*\|^2 = \underbrace{\|\mathcal{A}f - f^*\|^2}_{\textit{equivariant error}} + \underbrace{\|(I - \mathcal{A})f\|^2}_{= r^2\|\mathcal{A}f\|^2}. \quad (9)$$

Hence (i) *the squared error that any equivariantisation of f can remove is at most $r^2\|\mathcal{A}f\|^2$, and $\mathcal{A}$ attains it;* (ii) *the remaining error $\|\mathcal{A}f - f^*\|$ is equivariant error, which no symmetrisation of f reduces; and* (iii) *every map sending f to an equivariant field moves it by* at least $\|f - \mathcal{A}f\|/\|f\| = r/\sqrt{1+r^2}$*, with equality only for $\mathcal{A}$.*

Part *(iii)* is the direction the projection property actually gives, and it is worth separating from *(i)*: $\mathcal{A}f$ is the *nearest* equivariant field, so symmetrising is a lower bound on how much f is disturbed, never an upper one. What is bounded above is the *error* that symmetrising can remove, and only because the target is equivariant. Conflating the two — as an earlier version of this corollary did — reverses the inequality.

The practical reading is narrower than a decision rule, and worth stating as such. Measure r; if it is small, then by Eq. (9) the model's error is overwhelmingly *equivariant* error, and symmetrising f — by frame averaging, or

Arm (250 environments)	r under SE(3)	Free (%)
World frame	0.0881	15.4
World frame + SE(3) augmentation	0.0181	17.5
$(\mathbf{s}, \mathbf{g})$ reduction	0.0174	**45.6**

Table 1. The residual does not predict what a mechanism is worth. Augmentation and canonicalisation leave the learned field equally equivariant, and differ by 28 points of held-out performance

by any other projection onto the equivariant subspace — can remove at most an r^2 fraction of the field's squared magnitude from it. At $r = 0.016$ that is 2.5×10^{-4}, while the model is failing on more than half of its samples: the two quantities are not the same order, and no rearrangement of the symmetry can close the gap. The measurement costs one forward pass and no retraining.

What it does not license is a verdict on building an equivariant architecture, and on this point we can do better than a caveat. Symmetrising a trained field and training a constrained model are different operations: the first is a projection, bounded by Eq. (9); the second changes the hypothesis class, and with it optimisation, capacity allocation and generalisation, none of which r measures.

To test the tempting reading directly we measured r under the *full* SE(3) action rather than the roll alone, which makes it defined for the world-frame arm as well: K random rigid motions are applied to the whole problem, the field is evaluated in each, and each result is mapped back by the inverse rotation. Table 1 is the result.

Augmentation drives the residual down by a factor of 4.9: it demonstrably teaches the network the symmetry, to the same degree the reduction enforces it by construction. It buys 2.1 points. The reduction, at an indistinguishable residual, buys 30.3. *Two models with the same r can differ by 28 points*, so r does not forecast the benefit of a symmetry mechanism, and a decision rule of the form "build the constrained architecture only if r is large" has no support here. What r bounds is what symmetrising a given field can remove, which is Eq. (9) and is a theorem; what it does not measure is whether symmetry is the binding constraint at all.

This is also a consistency check on Proposition 1 for a *trained* model. For the reduced arm the SE(3) residual is 0.0186 at K=9 and 0.0194 at K=32, against 0.0191 and 0.0192 for the roll residual at matched K: a rigid motion acts on the reduced representation only through a roll, as claimed, and the two independently computed quantities agree to 0.0005.

We state the limit of this argument explicitly, because it is the objection a reader should raise. Corollary 5 bounds what can be gained by symmetrising *this* model. It does not bound what an equivariant architecture could achieve, since such an architecture searches a different hypothesis class and may converge to a different — possibly better — equivariant field. What a small r does establish is that the model's residual error is overwhelmingly *within* the equivariant subspace, so an architecture that merely enforces membership of that subspace is addressing a constraint that is already satisfied.

E Implementation

Three changes make the reduction representable, and one class of error had to be designed out.

Oriented boxes. The reduction is a general SO(3) rotation, under which an axis-aligned box becomes oriented. The usual six-number representation (centre, half-extents) cannot express this. We use twelve numbers: a centre and three half-edge *vectors*, which in the world frame are $(h_x, 0, 0)$, $(0, h_y, 0)$, $(0, 0, h_z)$, so the re-encoding is lossless. The environment SDF is unchanged, since collision checking happens in the world frame after the inverse transform.

Points and free vectors transform differently. This is the error the implementation is principally designed to prevent, because it fails silently. Waypoints, endpoints, and obstacle centres are *points* and take the affine map $R(\mathbf{p} - \mathbf{o})$; the box half-edges are *free vectors* and take the rotation only, $R\mathbf{v}$. Applying the affine map to a half-edge adds $-R\mathbf{o}$ to every edge — identically — so box *position* survives while box *extent* is swamped by a common offset. The model goes blind to obstacle size while the loss curve stays healthy. The severity scales with $\|\mathbf{o}\|$, so the defect is invisible on exactly those problems whose endpoints straddle the origin.

Verification. Eighteen tests cover deliberately disjoint failure classes. Asserting $\mathbf{s}, \mathbf{g} \mapsto (\mp d/2, 0, 0)$ catches translation errors, axis ordering, and R versus $R^\top$, but is blind to handedness, since a reflection in the y–z plane fixes the x-axis pointwise; $\det R > 0$ covers that. Neither detects the point/vector confusion, so a third test asserts $\text{featurise}(\rho \cdot S) = \text{typed\_rotate}(\text{featurise}(S), \rho)$ with the reference side written as explicit loops rather than the batched operations under test, so it catches indexing and reshaping errors instead of agreeing with them. A separate suite verifies Eq. (7) against a deliberately *untrained* network — the hard case, since Proposition 4 should not depend on f.

One subtlety is worth recording for anyone measuring r. The group acts on the whole input, scene and state together. An early version of our equivariance test rolled only the state while leaving the conditioning in its original frame, and reported a large violation from a correct implementation. A diagnostic written that way manufactures the non-equivariance it claims to find.

V Experiments

A Protocol

All results use environments 250–299, which no model was trained on, evaluated as 50 environments × 10 start–goal pairs × 20 samples with 8 Euler steps. The two arms are:

- **Control**: world frame, conditioned on the raw pair $(\mathbf{s}, \mathbf{g})$.
- **Treatment**: the $(\mathbf{s}, \mathbf{g})$ reduction with uniform roll augmentation, conditioned on the invariant scalar d.

Both arms use the twelve-dimensional oriented-box representation, so it is not a confound; both train on identical trajectories for 20 epochs.

Validation split. Within training environments, validation holds out whole start–goal pairs. A per-trajectory split leaks badly here: the 30 trajectories per pair are near-duplicates (within-pair standard deviation 0.054 against an overall 0.426, because CHOMP is a local optimiser and converges to the same route from similar initialisations). Under a per-trajectory split a held-out trajectory has a training neighbour at RMS distance 0.030; under the pair split this rises to 0.162.

Two success metrics. A sample is *free* if the whole path is collision-free. We report the per-sample rate throughout, because it is the quantity the ablations compare and it is not inflated by drawing more samples. The deployment metric is different: a generative planner is run as best-of-N behind a collision check, so what a user experiences is the fraction of *queries* for which at least one of N samples is free. We report that separately in Sec. V-B, where it can be compared against a baseline that is also stochastic. Mixing the two is the main way this kind of result gets overstated: at $N = 20$ they differ by tens of points.

Statistics. We treat the number of *distinct problems* (500) as the sample size rather than the number of trajectories (10,000), since samples within a problem and problems within an environment are correlated. This gives a standard error of $\approx$ 2 percentage points, and $\approx$ 2.7 for a difference of two arms. We checked that assumption rather than asserting it: a bootstrap that resamples whole *environments* — the outermost correlated unit — gives a standard error of 2.2 points on the 15.6% straight-line rate and 2.9 on the 74.0% TrajOpt rate (10,000 resamples), so the nominal figure is if anything slightly conservative at the rates that matter here.

Loss is not comparable across arms. The treatment regresses targets in the reduced frame, where trajectories are canonicalised along $+\hat{\mathbf{x}}$, so its target variance and hence its MSE are mechanically smaller. All reported quantities are world-frame task metrics.

B Calibration: floors and ceilings

A collision-free rate on a new benchmark means nothing without a scale, so we built one. Table 2 and Fig. 1 report four classical planners and a trivial baseline on the identical 500 problems, using the same collision checker. Classical planners are given one query at a time and run single-threaded on one CPU core.

Planner	Free (%)	Clearance	s/query
straight line	15.6	0.055	< 0.001
TrajOpt	74.0	0.026	0.352
CHOMP	88.8	0.019	0.394
RRT-Connect	99.6	0.017	0.284
expert (RRT+CHOMP)	100.0	0.037	0.342
flow, world frame	15.4	-0.070	0.067
flow, $(\mathbf{s}, \mathbf{g})$ reduction	**45.6**	-0.010	0.067

Table 2. The same 500 held-out problems, solved five other ways. Classical timings are one CPU core and one query; the learned rows are one GPU and 20 samples per query, so the time column is not a like-for-like comparison. Classical clearance is averaged over solved problems; the learned rows are per-sample rates at 250 training environments

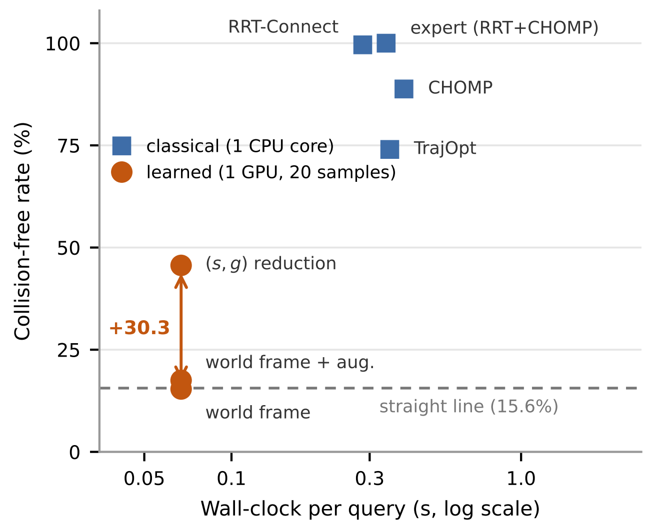

Figure 1. Where the learned planner actually sits. The world-frame arm lies on the straight-line floor; the reduction lifts it by 30.2 points; the classical planners remain far above. We show this rather than omit it: the contribution here is the representation result, not a competitive planner

Two facts frame everything that follows.

The floor. A straight line from start to goal is collision-free on 15.6% of these problems — the clutter is dense, but not so dense that the direct path is usually blocked. The world-frame model reaches 15.4% per sample. The two are within 0.2 points, an order of magnitude inside the standard error. On a per-sample basis, whatever the world-frame arm has learned is worth nothing over connecting the endpoints with a line. This is the sharpest available statement of the problem the reduction solves, and it is invisible without the trivial baseline.

Augmentation is not the same lever. The comparison above shows what *ignoring* the symmetry costs. The comparison a practitioner who already knows about the symmetry would want is against the other standard way of using it: augmenting the training distribution with random rigid motions of the whole problem. We ran that arm at all four scales (Fig. 2, Table 4). It helps, it helps more as data grows — augmentation needs data to exploit — and it is not close: 12.7, 13.8, 15.3, 17.5

Prior width σ	0	0.05	0.10	0.30
per-sample free (%)	15.6	11.5	7.4	0.4
best-of-20 (%)	15.6	**15.8**	13.4	2.2
mean path length	1.79	3.31	5.86	16.85

Table 3. The untrained stochastic baseline: Brownian bridges pinned at the endpoints, 20 samples per query, same collision check. Widening the prior buys nothing at any setting, so best-of-N over random detours is not what a learned sampler is competing against

against the reduction's 19.0, 34.4, 41.0, 45.6. At the largest scale augmentation captures about 7% of what canonicalisation does, and leaves the world-frame arm within 2 points of the straight-line floor. Making equivariance *likely* through the data is a different and much weaker mechanism than removing the degrees of freedom outright.

The stochastic floor. A straight line has one sample, so it cannot be compared against a best-of-N number. The honest control there is the model's own prior with the learning removed: Brownian bridges pinned at start and goal, drawn $N = 20$ times, filtered by the same collision check. We sweep the prior width and report the *best* setting, so the baseline is steel-manned (Table 3). Perturbing the straight line does not help. The best best-of-20 result over the whole sweep is 15.8% at $\sigma = 0.05$, against 15.6% for the unperturbed line: random detours leave the workspace faster than they route around anything, and by $\sigma = 0.3$ the mean path is nine times longer than the direct one and solves 2.2%. The $\sigma = 0$ row also reproduces the straight-line rate to the digit through an independent code path, which is a check on both harnesses.

This changes the reading of the world-frame result in a way worth stating carefully. Per sample, that arm is at the straight-line floor. Per query at $N = 20$ it is not: the world-frame model trained on 20 environments solves 44.8% of queries with at least one free sample, against 15.8% for the best untrained prior at the same N. So the world-frame model has learned something real — its samples are diverse in an informative way rather than a random one — but it has not learned to make any *individual* sample better than a straight line. The reduction is what closes that second gap.

The ceiling. RRT-Connect solves 99.6% and the expert pipeline 100%, in under 0.4 s of single-core CPU time; CHOMP from a straight-line initialisation solves 88.8%. The learned sampler at 45.6% is not competitive with any of them. We report this plainly because the alternative — comparing a learned planner only against other learned planners — is how a benchmark stops measuring anything. The contribution of this paper is that the representation change is worth 30 points and the symmetry machinery is worth none; that contribution does not require the planner to win, and we do not claim it does.

Training	Collision-free (%)			best-of-20
envs	World	+aug.	Reduction	Reduction
20	12.8	12.7	19.0	57.8
60	13.6	13.8	34.4	79.6
150	14.6	15.3	41.0	85.6
250	15.4	17.5	**45.6**	**90.4**

Table 4. Held-out performance against training-set size. "+aug." is the world-frame arm trained with random SE(3) motions of the whole problem. The world-frame arm gains 2.6 points across a 12.5× increase in environments and augmentation adds at most 2.1 more, while the reduction gains 26.6 and has not plateaued. The 60-environment row is a mean over three seeds; the rest are single seed. The last column is the deployment metric, best-of-20 behind a collision check

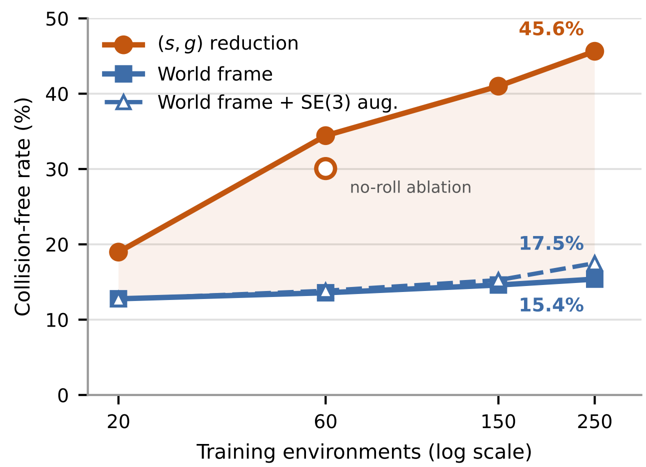

Figure 2. The gap widens with data. If canonicalisation were compensating for a shortage of training data one would expect it to close; instead the world-frame baseline is nearly flat while the reduced model continues to improve. The hollow marker is the no-roll ablation, drawn on the treatment line because it is the same arm with one component removed

C The reduction scales; the baseline does not

Table 4 and Fig. 2 give the principal result. At 250 environments the reduction reaches 45.6% against 15.4%, a gap of 30.3 points. Measured against the across-seed standard error at 60 environments (1.56 points for this arm, n=3), that is roughly 19 standard errors; the seed spread was not measured at 250 environments, so we quote the 60-environment floor, which is the wider of the two arms.

The trend is more informative than the endpoint. The baseline improves by 2.6 points across a 12.5× increase in training environments: it is close to unable to exploit additional layouts, because each new layout arrives at poses it must learn about separately. The reduction improves by 26.6 points over the same range and has not levelled off. This is the opposite of what a small-data-crutch account predicts, under which the advantage should shrink as data accumulates. Minimum clearance corroborates through an independent continuous measure: the treatment moves from −0.058 toward −0.0095, approaching

Mechanism	Δ free	Assessment
$(\mathbf{s},\mathbf{g})$ reduction (5 DOF)	+30.3 pp	$\approx$ 19 SE; grows
SE(3) augmentation (world frame)	+2.1 pp	real, but 7% of the above
Roll augmentation	+4.4 pp	2.8 SE; one seed, see text
Frame averaging (K:1→3)	+0.1 pp	0.05 SE ($\approx$ 0)

Table 5. Decomposition by mechanism on the same held-out set. Among the mechanisms tested, canonicalisation accounts for the effect: the next largest is an order of magnitude smaller, and the two built for the sixth degree of freedom are smaller still. The Δ column is against the arm without that mechanism, so the rows are not additive

feasibility, while the baseline is static.

Endpoint error falls by a factor of three to five under the reduction ($0.0101 \to 0.0041$ at 20 environments, $0.0059 \to 0.0020$ at 250), a direct consequence of canonicalisation: with endpoints always at $(\mp d/2, 0, 0)$ the model no longer spends capacity on *where* they are.

D The mechanisms for the sixth degree of freedom are inert

Table 5 and Fig. 3a decompose the result. Removing roll augmentation (the reduction with a fixed gauge) costs 34.4% → 30.1% at 60 environments, on which see the caveat in Sec. VI. Frame averaging on the best model gives 45.63% at $K{=}1$ and 45.70% at $K{=}3$ — 0.05 across-seed standard errors.

Rather than infer inertness from flat curves, we measure r. On the best model $r = 0.0168$ at $K{=}3$: under 2% of the field is non-equivariant. By Eq. (9) the squared error any equivariantisation can remove is at most r^2 of the field's squared magnitude, i.e. about 2.5×10^{-4}, while the model is failing on more than half of its samples. The remaining error respects the symmetry and is untouchable by symmetrisation.

Two observations sharpen this. First, removing roll augmentation raises the residual by 64% ($0.0170 \to 0.0279$ at 60 environments) — a large relative change, but between two values that are both under 3%, so augmentation is not what makes the field near-equivariant in the first place. The explanation is that the gauge is a deterministic function of $\hat{\mathbf{x}}$, and 600 start–goal pairs per environment supply 600 distinct directions; the reduced-frame data already spans a wide spread of rolls, and explicit augmentation was solving a problem the data had already solved. Second, and contrary to what we expected, the residual does *not* move with scale: 0.0183, 0.0170, 0.0177, 0.0168 across a 12.5× range of training data (Fig. 3b). It is small and flat. An earlier two-point reading under a mean-of-norms definition of r suggested a downward trend; measured at four scales under Eq. (8) it is a constant, and we report it as one.

That the residual is flat rather than falling is itself informative. It means the near-equivariance of the learned field is not something the model acquires gradually from seeing more layouts — it is present at 20 environments and unchanged at 250. Combined with the no-roll ablation, the picture is that the reduced-frame data supplies enough implicit roll diversity from the outset, and that nothing about the scale of the training set changes how equivariant the learned field is. The case for not building a constrained architecture rests on the *size* of r and Corollary 5, neither of which depends on a trend.

E How much of the level is training budget

Every arm above is trained for 20 epochs, and that budget is not enough. Holding the data fixed at 60 environments and training the reduced arm for 60 epochs instead raises the held-out collision-free rate from 31.3% to **42.4**%, with minimum clearance improving from −0.031 to −0.013 and endpoint error from 0.0024 to 0.0022. Three times the optimisation, at identical data and identical capacity, is worth +11.1 points — an order of magnitude more than either mechanism built for the sixth degree of freedom.

Two consequences follow, and the second is a correction.

First, this is the direct evidence that the absolute level is a budget artefact rather than a ceiling of the method, and it is the evidence we rely on rather than the train–validation comparison discussed in Sec. VI. The model at 20 epochs is substantially underfit, so the 45.6% headline understates what this architecture reaches at convergence.

Second — and this is the part we did not anticipate — the same tripling is worth almost nothing to the world-frame arm. Running it at 60 epochs on the same 60 environments moves it from 13.6% to 14.4%:

60 environments	20 epochs	60 epochs
World frame	13.6	14.4
$(\mathbf{s},\mathbf{g})$ reduction	34.4	**42.4**

Optimisation buys +8.0 points with the reduction and +0.9 without it, an interaction of +7.1 points at roughly 4.5 across-seed standard errors. The two mechanisms are not substitutes: they relieve different constraints. The representation bottleneck — the network being shown a pose it must relearn at every value — is removed by canonicalisation and *cannot* be removed by compute. The capacity bottleneck is removed by compute and cannot be removed by canonicalisation. This also settles the reading of Fig. 2: the world-frame arm's flatness is not an artefact of an insufficient budget, because giving it three times the budget leaves it flat.

The deployment metric at this budget is high: best-of-20 behind a collision check solves **87.2**% of held-out queries, against 15.8% for the best untrained prior at the same N (Table 3). A reader who cares about whether the sampler is useful rather than about the representation question should read that number, with the caveat from Sec. V-B that classical planners still solve more of them and faster.

F Cost

Frame averaging at $K{=}9$ costs 1.05× the wall-clock time of $K{=}1$ (0.070 vs 0.067 s per query). At 64 waypoints

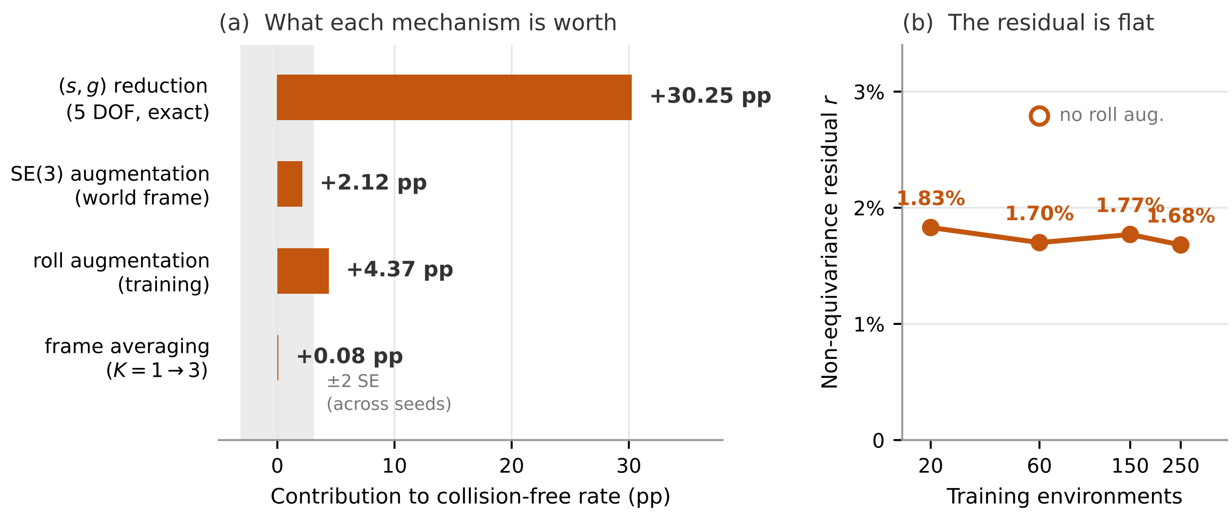

Figure 3. (a) Only the canonicalisation clears the noise floor; the two mechanisms built for the sixth degree of freedom sit inside it. (b) The non-equivariance residual of Eq. (8) is smaller at the larger scale — two points at one seed, so read it as consistent with the property arriving on its own rather than as establishing it; the hollow marker shows that removing roll augmentation raises the residual but leaves it small, so augmentation was not the cause of the near-equivariance

and batch size 1 the sampler is bound by kernel launch overhead rather than arithmetic, so a ninefold increase in width is nearly free. This matters for the interpretation: the finding that frame averaging does not help is not a budget trade-off. It was affordable, and we adopted it anyway for the measurement.

Separately, reducing the integrator from 20 to 8 Euler steps costs 0.4–0.8 points and runs 1.4× faster, so 8 steps is used throughout.

VI Discussion and Limitations

What the technique contributes. Of the six degrees of freedom of SE(3), five can be removed exactly by a frame read off the query, at no computational cost and with no architectural constraint. Doing so roughly triples the collision-free rate on this benchmark and, more importantly, converts a model that cannot use additional training environments into one that can. The sixth cannot be removed and, here, does not need to be — a contingent empirical fact rather than a theorem, but one supported by three independent measurements: the ablation, the K sweep, and the residual.

The learned planner loses to the classical ones. Table 2 is unambiguous: RRT-Connect solves essentially every problem in 0.28 s of single-core CPU time, and the best learned arm solves 45.6% per sample. Part of this is budget — the model is small (2.16M parameters) and briefly trained (20 epochs) — and part is that a 64-waypoint unconditional-prior sampler is simply a weaker instrument than a complete planner on a problem with an exact SDF. We draw the comparison anyway, because a representation result that is only ever measured against other configurations of itself is not anchored to anything.

We are careful about how the underfitting claim is supported, because the obvious support is confounded. Comparing the epoch-averaged training loss against a validation loss computed from the exponential-moving-average weights is not a generalisation gap: the training figure lags, since it averages over an epoch during which the weights were still improving, and the two figures are computed from different weights, with the EMA weights typically the better ones. Both effects push validation below training regardless of fit. We therefore score a fixed slice of the training set with the same EMA weights and the same code path as validation, and read the gap only from that like-for-like pair. The direct evidence is the longer run of Sec. V-E: at fixed data and fixed capacity, three times the optimisation is worth +11.1 points of held-out collision-free rate, which is what underfitting looks like measured on the task rather than on the loss.

The claim we make is comparative and holds at every point of the scaling curve; the claim we do not make is that this planner should be used.

Scope of the negative result. The claim that the sixth degree of freedom needs no treatment is specific to this setting, and relies on the training distribution supplying implicit roll diversity through many start–goal directions. A dataset with few directions per environment, or a task whose angular structure is genuinely richer — manipulation with a gripper, or non-holonomic dynamics — could plausibly reverse it. That is precisely why we advocate measuring r rather than generalising our answer: the diagnostic transfers even where the conclusion does not.

What the residual does and does not bound. As stated in Sec. IV-D, Corollary 5 only bounds post-hoc symmetrisation of a given model, not the achievable performance of a differently-parameterised equivariant model. A conclusive test would train an SO(2)-

equivariant backbone and compare; our argument is that an r this small makes that experiment low-yield, not that it is uninformative.

One honest asterisk. Frame averaging *does* consistently improve endpoint error, by roughly 35% ($0.0020 \to 0.0013$), on every checkpoint tested. This is a real effect. It simply does not propagate to the collision-free rate, because endpoint precision was not the failure mode.

Dataset diversity. The dataset is less diverse than its nominal size suggests: 30 near-duplicate trajectories per pair means 10.8M paths carry on the order of 360k distinct queries. This does not affect the comparison, since both arms train on identical data, but it bears on how additional data should be generated — cutting trajectories per pair from 30 to 5 would buy roughly six times the environments for the same generation cost. We no longer read Fig. 2 as saying environments are the single binding constraint, since Sec. V-E shows optimisation buys a comparable amount at fixed data; the honest statement is that both are binding and neither has been pushed to saturation.

One arm the comparison still needs. A translation-only reduction, which recentres on the midpoint without rotating, would split the 30.3 points between the three translational and the two rotational degrees of freedom; we report them only in aggregate. It is one training run against the existing pipeline.

The roll-augmentation ablation needs seeds. With three seeds the reduced arm at 60 environments averages 34.4%, against 30.1% for the same arm without roll augmentation — +4.4 points, or about 2.8 across-seed standard errors. That is larger than the +1.2 we would have reported from single seeds, and it is no longer obviously inside noise. But the no-roll ablation itself has one seed, so this compares a single draw against a three-seed mean and we do not draw a conclusion from it. Given the treatment arm's seed spread (31.3, 36.0, 35.9), single-seed ablations at this scale are not reliable, which is a caution that applies to the K sweep as well — though there the effect is 0.05 SE rather than 2.8.

One generative objective, one seed. Every learned number here comes from conditional flow matching at a single training seed. The reduction is a statement about the problem rather than about the interpolant, so it should transfer unchanged to a denoising-diffusion objective of the Diffuser/MPD family [12, 3]; we have not measured that, and until it is measured the possibility that the effect is specific to straight-line interpolants cannot be excluded. Likewise, the reported standard errors come from the spread over held-out problems, not over training seeds, so they do not capture run-to-run variance in optimisation. Both are cheap to close and both belong in a camera-ready version.

Single benchmark. Results are from one workspace family and one planner architecture. We report the reduction and the diagnostic in enough detail to be reimplemented elsewhere; whether the 30-point gap survives in a manipulator configuration space is open.

Reproducibility. The benchmark, the expert pipeline, both learned arms, the classical baselines, and the residual diagnostic are one codebase; every number in this paper is produced by a script that takes the checkpoint and the held-out shard range as arguments, and the figures are regenerated from those outputs rather than transcribed. The equivariance and geometry test suites described in Sec. IV-E run in seconds and are the intended entry point for anyone reimplementing the reduction, since they encode the failure modes that are silent at training time.

Future work. Beyond more environments and capacity, one option is now open that was previously excluded. The flow begins from an isotropic Gaussian that knows nothing of the query, so it must learn both to produce a path and to place its endpoints. A Brownian-bridge prior pinned at start and goal would leave only the obstacle-avoidance detour to be learned. We had excluded it because such a prior is not isotropic in the plane normal to $\hat{\mathbf{x}}$ and would break the distributional equivariance guarantee at $t = 0$; since that guarantee has now been measured to be worth 0.2 points, the trade is available.

VII Conclusion

The model presented here is exactly SE(3)-equivariant by construction for five of six degrees of freedom, and measured to be 98% equivariant for the sixth without any mechanism enforcing it. All of the measured benefit lies in the five.

The recommendation is correspondingly simple, and we believe it applies beyond this benchmark: canonicalise the degrees of freedom that admit a continuous canonical form — the query usually supplies more of them than one expects — and then *measure* the residual non-equivariance rather than assuming it. The measurement costs one forward pass.

What that measurement licenses is worth stating exactly, since it is narrower than it is tempting to write. It says that *symmetrising this model* is worth almost nothing here, because almost none of this model's error is a symmetry violation. It does not say that a network built equivariant from the start would fail, since such a network searches a different hypothesis class and could converge somewhere better. The claim is about where the error lives, not about what a different architecture could achieve. Turning r into a genuine model-selection rule would require showing that it predicts the gain from constrained architectures across tasks with differing amounts of non-equivariance, including tasks where the symmetry genuinely fails; that is future work, and we state it as such rather than implying it here.

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